Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Data Platform · Big Data · Apache Spark · Kubernetes · Observability · Python · Java · Scala · AWS

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SUMMARY

Software engineer with 9+ years building and operating production data platforms and backend services end-to-end, on-prem and cloud. At Entrata (7+ years, 100K+ users), owned mission-critical ETL data pipelines, async message bus processing (RabbitMQ), database optimization, and AWS infrastructure across a regulated multi-tenant SaaS platform. At Intuit, building ML data pipelines and Python/FastAPI backend services with observability and CI/CD on AWS. Strong across the full data platform stack: data modeling, data transformation and warehousing (Snowflake), Apache Spark/PySpark, Kubernetes microservices, message buses (Kafka, RabbitMQ), API gateways, database abstraction layers, and end-to-end observability with OpenTelemetry, Prometheus, and Grafana. Built AgentProbe - a data platform with tracing, structured logging, and telemetry across distributed service runs. Python, Java, Scala, TypeScript; PostgreSQL, Redis, Snowflake, SQLite; AWS (S3, RDS, Lambda, Glue, SQS). Owner's mindset: delivered complex systems on schedule with zero compliance violations across 7 years. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Data Pipelines · ETL/ELT · Data Transformation | Apache Spark · PySpark · Big Data Technologies | Data Warehousing · Snowflake · Data Modeling | Message Buses · Apache Kafka · RabbitMQ | API Gateways · REST APIs · Database Abstraction Layers | Kubernetes · Docker · Microservices | Observability · Tracing · OpenTelemetry | Prometheus · Grafana · Telemetry | Python · Java · Scala · TypeScript | PostgreSQL · Redis · MySQL · Oracle | AWS (S3 · RDS · Glue · Lambda · SQS) | Database Optimization · Query Performance · Indexing | CI/CD · GitHub Actions · Automated Testing | SaaS · Agile/Scrum · On-Prem & Cloud | Owner's Mindset · Cross-Functional Collaboration

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA

- Build ML data pipelines in Python (scikit-learn, XGBoost) from data ingestion and feature engineering through model evaluation and prediction APIs; develop backend Python/FastAPI services with validation, error handling, and observability on AWS.
- Design structured-output schemas for LLM-powered financial data summarization; build evaluation and regression test pipelines for AI outputs; containerize services with Docker through GitHub Actions CI/CD to AWS with logging and monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **Owned the HUD income-limit ETL data pipeline for 6 years** - complex data transformation ingesting federal source rows into ~180,000 normalized domain records powering eligibility, rent, and compliance calculations for 100K+ housing units. Validation, reconciliation, and rollback preparation every release cycle. Zero compliance violations across the full ownership period.
- **Built and maintained RabbitMQ-based async message bus processing** for rent updates across thousands of tenant records - designed retry logic, idempotency guarantees, and failure-routing that eliminated double-processing incidents; owned the system through production incidents and annual schedule changes.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant schemas for eligibility, financial, and document workflows.
- Docker microservices redesign (50% scalability improvement, 30% faster deployments); PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage, zero incidents); Python/Bash automation eliminating 80% of manual operational workload; mentored 5+ engineers; 300+ code reviews.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data import and reconciliation workflows (1.6M records); led AWS S3 cloud migration.

PROJECTS

AgentProbe - Distributed Telemetry, Tracing & Observability Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI data platform with structured logging, full tracing across distributed service runs, telemetry collection, and analytics pipeline - ingests run data, transforms it through a 5-metric scoring engine, warehouses results, and surfaces quality metrics via CI-integrated regression harness. Multi-provider routing with cost and latency tracking per run.

Leasel - Enterprise Data Platform with Compliance ETL & API Layer

2026 | Python, FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end data platform: ingestion pipeline (OCR + document parsing), data transformation and normalization, hybrid search retrieval layer, FastAPI serving with defined API contracts, and CI regression gate. Three consuming surfaces share one data warehouse layer. 93.6% precision across 1,240 test cases.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 — May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017